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From Planetary Diagnostics to Collective Response,Diagnostic-to-Actuation Coupling as the Actuation Layer of the Emergent Mind Observatory

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Abstract:	Earth system science, systemic-risk research, digital Earth twins, and early-warning systems have greatly improved the detection and modelling of planetary stress. What remains weakly formalized is the conversion of diagnostic capacity into organized collective response. This article argues that diagnostic-to-actuation coupling (DAC) is a missing variable in futures-oriented governance: the strength with which validated signals of danger are translated into coordinated policy, budgetary commitment, institutional delivery, and durable collective behavior. It positions DAC as the actuation layer of the Emergent Mind Observatory (EMO), linking planetary diagnostics, shared self-models, and behavioral response. The framework analyzes near-future human response through six variables - organismality, integrative processing, global workspace ignition, self-model fidelity, DAC, and total stress load - and derives five recurrent response modes. The contribution is methodological and anticipatory rather than metaphysical: DAC is formalized as a distinct governance variable within an operational EMO architecture, differentiated from adjacent metrics, and used to infer patterned collective response under planetary stress.

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Dear Editors,

Please consider the enclosed manuscript, “From Planetary Diagnostics to Collective Response: Diagnostic-to-Actuation Coupling as the Actuation Layer of the Emergent Mind Observatory,” for publication as a Full Length Article in Futures.

This article argues that diagnostic-to-actuation coupling (DAC) is a missing variable in futures-oriented governance: the strength with which validated signals of danger are translated into coordinated policy, budgetary commitment, institutional delivery, and durable collective behavior. The manuscript positions DAC as the actuation layer of the Emergent Mind Observatory (EMO), linking planetary diagnostics, shared self-models, and patterned behavioral response under planetary stress.

The contribution is methodological and anticipatory rather than metaphysical. The article formalizes DAC as a distinct and measurable governance variable, embeds it within an operational EMO architecture, differentiates it from adjacent concepts such as policy coherence, early-warning assessment, adaptive governance, and implementation science, and shows how it can support behavioral inference across recurrent response modes. In doing so, the manuscript speaks directly to Futures’ interest in anticipation, foresight, governance, and long-range human response to systemic risk.

This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. The submission has been prepared for double-masked review. The author declares that there are no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

I appreciate your consideration.

Sincerely,

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Title

From Planetary Diagnostics to Collective Response: Diagnostic-to-Actuation Coupling as the Actuation Layer of the Emergent Mind Observatory

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Highlights

- The article formalizes DAC as a measurable governance variable.
- DAC is positioned as the actuation layer within EMO.
- The framework links validated warning to policy, budgets, and delivery.
- Six variables support inference across recurrent response modes.
- The model strengthens anticipatory governance under planetary stress.

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From Planetary Diagnostics to Collective Response

Diagnostic-to-Actuation Coupling as the Actuation Layer of the Emergent Mind Observatory

Abstract

Earth system science, systemic-risk research, digital Earth twins, and early-warning systems have greatly improved the detection and modelling of planetary stress. What remains weakly formalized is the conversion of diagnostic capacity into organized collective response. This article argues that diagnostic-to-actuation coupling (DAC) is a missing variable in futures-oriented governance: the strength with which validated signals of danger are translated into coordinated policy, budgetary commitment, institutional delivery, and durable collective behavior. It positions DAC as the actuation layer of the Emergent Mind Observatory (EMO), linking planetary diagnostics, shared self-models, and behavioral response. The framework analyzes near-future human response through six variables - organismality, integrative processing, global workspace ignition, self-model fidelity, DAC, and total stress load - and derives five recurrent response modes. The contribution is methodological and anticipatory rather than metaphysical: DAC is formalized as a distinct governance variable within an operational EMO architecture, differentiated from adjacent metrics, and used to infer patterned collective response under planetary stress.

Keywords: futures studies; anticipatory governance; Earth system governance; collective intelligence; diagnostic-to-actuation coupling; planetary stress

1. Introduction

Humanity has become powerful enough to alter the Earth system, yet it still lacks a mature science of its own collective behavior. Earth-system research can now observe climate, oceans, land, ice, biospheric instability, and compound hazards with extraordinary precision. Governance research can study institutions, incentives, and policy designs. Risk science can map cascades, feedbacks, and polycrisis dynamics. What remains underdeveloped is a compact framework for

asking how the species itself behaves when multiple stressors intensify at once - and how strongly knowledge of danger is translated into coordinated adaptation.

This gap matters because the defining crises of the twenty-first century are not isolated. Climate destabilization, war, hunger, migration stress, inequality, ecological overshoot, and information overload interact. Humanity has built extraordinary capacities for sensing, modelling, and warning, but these do not automatically yield coherent action. A civilization may detect danger, simulate danger, and still respond incoherently. It may know more and do less. The central scientific problem is therefore no longer only whether danger can be measured. It is whether response behavior can be measured as well. For anticipatory governance, the decisive question is not only whether futures are modeled but whether validated futures meaningfully alter present-day trajectories.

Diagnostic-to-actuation coupling is defined here as the strength with which validated signals of danger are translated into synchronized policy, budgetary commitment, institutional delivery, and durable collective behavior. The paper places DAC within the architecture of the Emergent Mind Observatory as its functional response layer. EMO provides the observatory, the shared self-model, and the cross-domain measurement architecture. DAC measures how strongly the outputs of that observatory are converted into organized response.

This move matters analytically. Once DAC is embedded within EMO, the framework becomes less metaphorical and more instrument-like. The question is no longer simply whether humanity resembles a macro-organism in some philosophical sense. The question becomes whether species-scale sensing, modelling, and anticipatory capacity can be linked to measurable actuation pathways. In this sense, the paper moves from a general diagnosis of the warning-to-action gap to a structured architecture for observing that gap, comparing it across domains, and inferring the response modes that follow from it. The contribution of this article is to formalize DAC as a measurable actuation variable within EMO and to show how it enables behavioral inference under planetary stress.

2. The unresolved gap between planetary diagnostics and organized response

Contemporary research is strong in at least three areas. First, Earth-system science has clarified the scale of planetary destabilization and the concept of safe and just boundaries. Second,

systemic-risk and polycrisis research has shown that threats interact across domains rather than arrive one by one. Third, early-warning systems have expanded the ability to detect danger, especially for weather, climate, and disaster risk. Yet these gains leave a missing layer: the relation between validated diagnosis and durable collective actuation.

The persistence of this gap is visible across domains. Climate modelling becomes more precise while implementation remains late relative to trajectories. Disaster preparedness may improve at the level of detection and scenario planning while stockpiles, surge capacity, logistics, and cross-jurisdictional deployment remain weakly coupled to those warnings. Pandemic response demonstrated the same asymmetry: surveillance, dashboards, and scenario models can be abundant before procurement, distribution, and institutional coordination are stabilized. In each case, the civilization sees more clearly than it acts.

That asymmetry cannot be reduced either to general state capacity or to policy coherence alone. State capacity concerns the ability to execute once a decision is made. Policy coherence concerns alignment across goals, instruments, and behavior. DAC is narrower than the first and sharper than the second. It concerns the strength with which validated diagnostics are converted into cross-scale, budgeted, persistent actuation. It is therefore the most direct variable for measuring whether a civilization's sensory and modelling apparatus actually enters the machinery of collective response.

For futures research, this is not a secondary issue. If the future is increasingly shaped by compound stress and anticipatory infrastructures, then the decisive variable is not only what societies know, but how strongly their validated knowledge changes response trajectories. A futures-oriented governance framework that measures sensing without measuring conversion into action remains incomplete.

2.1. Relation to existing governance metrics

DAC is adjacent to several established governance constructs, but it is not equivalent to any of them. OECD policy-coherence frameworks are concerned with whether policies across sectors and time horizons work together, how trade-offs are managed, and whether institutional coordination is robust enough to support sustainable development. That literature is indispensable, but its primary object is alignment across policies rather than the conversion

strength by which validated diagnostics become organized response. DAC therefore presupposes some issues addressed by policy coherence while focusing on a different bottleneck: the transmission path from recognized danger to coordinated actuation.

The distinction from early-warning assessment is equally important. UNDRR and related work on people-centered, end-to-end, multi-hazard early-warning systems emphasize risk knowledge, monitoring, dissemination, preparedness, and the triggering of early action. DAC does not replace that architecture. It asks a narrower downstream question: once danger is institutionally validated, how strongly does that validation survive translation into policy instruments, budgetary commitments, operational delivery, and persistence after attention fades? In that sense, DAC begins where successful warning systems become behaviorally consequential.

DAC also differs from adaptive governance. Adaptive-governance research is primarily concerned with institutional learning, flexibility, polycentricity, and the capacity to adjust under uncertainty. Those questions concern the adaptive qualities of governance arrangements. DAC instead measures a specific conversion relation. A system may be normatively committed to adaptive governance and still exhibit weak warning-to-budget translation, weak alert-to-deployment conversion, or rapid post-ignition decay. Likewise, DAC is related to implementation science but is not reducible to it. Implementation science studies how policies or interventions are executed in real settings; DAC isolates a prior and broader relation at the planetary-governance level: whether validated diagnostics reliably enter the machinery of collective response across domains, scales, and time.

3. From EMO and UIA to DAC

The Emergent Mind Observatory can be understood as a planetary observatory for measuring coherence, contradiction, and emergent response patterns across the human-Earth interface. In that architecture, major data systems are treated as interfaces: Earth-system models and digital twins, event graphs, financial and trade networks, health and demographic systems, governance repositories, and early-warning infrastructures. UIA supplies the cross-domain ledger by organizing informational curvature, focusing, coherence, entropy production, information flow, and semantic efficiency into a common formal language.

That prior architecture already contains the ingredients DAC needs. EMO's species-mind layer includes organismality, synergy, global workspace ignition, self-model fidelity, and information-time. It also includes coupling between model outputs and actuation. The missing step is to isolate that coupling and treat it as a first-order explanatory layer rather than a subsidiary feature. In other words, DAC should not sit beside EMO as a useful conceptual extension. It should be understood as the point within EMO at which diagnostics become behaviorally consequential.

Placed this way, the relation between EMO and DAC becomes clearer. EMO measures the mirrors a civilization builds: what it senses, models, remembers, and broadcasts about itself and the planet. DAC measures how strongly the image in that mirror enters the muscles of collective action. EMO without DAC remains observational. DAC without EMO remains underspecified. Together they create a tractable architecture for studying whether humanity's growing self-observation is becoming operationally meaningful.

UIA strengthens this interpretation. At planetary scale, EMO's metric stack can already populate components of a_UIA and its coarse-grained counterpart. That matters because UIA gives EMO a common measurement language for curvature, coherence, information flow, and semantic efficiency rather than a loose bundle of parallel indicators. Within that shared ledger, DAC can be treated as the actuation bridge that tests whether changes in the self-model are behaviorally consequential at the level of policy, budgets, and durable coordination. DAC then appears as the decisive actuation bridge between self-model fidelity and durable response. The move proposed here is therefore consistent with the broader UIA program: the point is not to add metaphysical grandeur, but to identify the missing functional layer through which species-level cognition affects planetary behavior.

4. DAC as the actuation layer of EMO

DAC is best understood not as one scalar but as a layered conversion architecture inside EMO. Its function is to measure how validated warnings, risk models, and self-model outputs travel from diagnosis to organized response. Four sublayers are especially important.

First is diagnostic validation. Not all signals deserve actuation. DAC begins only when a signal has crossed a threshold of institutional and epistemic legitimacy: validated forecasting, recognized warning, or sufficiently corroborated model output. This distinction matters because

DAC is not a measure of reaction to noise; it is a measure of how strongly validated danger becomes organized response.

Second is translation. Once danger is recognized, the signal must be translated into policy language, budgetary commitments, operational mandates, and implementation pathways. Here DAC can be measured through warning-to-policy lag, model-to-budget alignment, and the degree to which institutional choices preserve diagnostic content rather than symbolically acknowledging it while stripping away operational force.

Third is deployment and persistence. Actuation is not exhausted by declaration. DAC must therefore include alert-to-deployment conversion rates, the continuity of implementation after media attention decays, the maintenance of logistics and infrastructure over time, and the degree to which cross-scale coordination remains intact under friction. This is where DAC diverges most clearly from rhetorical governance.

Fourth is contradiction detection and relapse monitoring. One of EMO's core strengths is that it can register contradiction between declared goals, policy instruments, budgets, events, and delivery. DAC must inherit that reflexivity. A civilization may seem to respond while actually generating cross-pressured, self-cancelling behavior. DAC therefore includes measures of contradiction, selective under-implementation, decay after initial mobilization, and relapse into fragmented subsystem behavior.

Taken together, these sublayers make DAC a functional response stack. EMO's lower layers register interfaces, ingest data, and compute shared metrics. DAC sits above those as the structured bridge between validated diagnostics and organized delivery. It turns EMO from an observatory of civilizational self-knowledge into an observatory of warning-to-action conversion.

5. Behavioral inference under planetary stress

Once DAC is embedded inside EMO, the move from diagnosis to behavioral inference becomes more precise. The framework retains six variables: organismality (O), integrative processing (I), global workspace ignition (G), self-model fidelity (M), diagnostic-to-actuation coupling (D), and total stress load (S). The forecasting logic can still be written compactly as $B(t) = f[O(t), I(t),$

G(t), M(t), D(t), S(t)], where B denotes the behavioral mode distribution of the partially integrated species.

Organismality measures the degree to which destructive conflict is suppressed and cooperative continuity is sustained across scales. Integrative processing measures whether heterogeneous subsystems combine information into better collective judgment. Global workspace ignition captures punctate episodes in which signals become system-wide objects of attention. Self-model fidelity measures the strength with which diagnostic representations actually guide collective orientation and intention. DAC measures the narrower but sharper question of whether validated danger is converted into coordinated, budgeted, and durable response. Stress load measures the aggregate burden imposed by ecological, geopolitical, metabolic, technological, and informational pressures.

The distinction between M and D is therefore sequential but empirically separable. M concerns internal cognitive alignment: whether the civilization's representations of itself and its risks are accurate enough, shared enough, and behaviorally salient enough to shape declared priorities. D concerns external operational alignment: whether those priorities survive contact with institutions and become organized delivery. Self-model fidelity concerns cognitive alignment; DAC concerns operational conversion. A system may know accurately yet act weakly. A system may possess a relatively strong self-model while still exhibiting weak actuation because warning is diluted in budgeting, mandate formation, logistical execution, or persistence. Conversely, strong episodic deployment can occur under a weak self-model when mobilization is reactive, fragmented, or strategically inconsistent. The variables are related, but they do not collapse into one another.

From these variables, five recurrent response modes follow. Delayed coordination appears when sensing and modelling are stronger than DAC and when the tempo of institutional conversion lags the tempo of the threat; a familiar pattern is the long interval between repeated climate warnings and binding implementation at the necessary scale. Pulsed adaptation appears when ignition is high but persistence is weak: the system briefly aligns, then decays, as in short-lived emergency mobilizations that fade once media attention recedes. Contradictory action appears when diagnostics are formally acknowledged but budgets, incentives, and infrastructures remain cross-pressured, for example when decarbonization targets coexist with durable fossil-fuel

subsidy or expansion logics. Selective stabilization appears in domains where organismality, self-model fidelity, and DAC rise together strongly enough to sustain coordinated response, as in targeted vaccination, flood-protection, or grid-resilience programs that remain institutionally supported after initial warning. Fragmentation relapse appears when temporary integration gives way to fatigue, rivalry, or renewed local optimization, as when an initially coordinated crisis response breaks back into jurisdictional competition, procurement fragmentation, or policy reversal.

These modes should not be treated as exhaustive. They are proposed as a first compact typology of recurrent species-level conduct under compound stress. Their analytic value lies in making behavior inferable from measurable properties of integration, ignition, coupling, and overload.

6. Empirical strategy and implementation pathway

The framework becomes useful only if it is falsifiable and implementable. Four outcomes would count against it. First, DAC might fail to improve forecasting beyond conventional sectoral indicators. Second, the relation between ignition, self-model fidelity, DAC, and response timing might prove weak or unstable. Third, the proposed behavioral modes might collapse empirically, proving too vague to distinguish. Fourth, DAC itself might fail to add explanatory value beyond already existing measures of governance performance.

The advantage of integrating DAC into EMO is that the minimum viable dataset is already largely specified. The coherence work points toward a declared goal vector, a policy-instrument layer, a budget-and-delivery layer, an event-response layer, and a validation layer. EMO already specifies the ingestion of multistream data and the computation of species-level cognition metrics. DAC uses that architecture but asks a more behaviorally direct question: how strongly do validated diagnostics become policy, budgets, deployment, and persistence?

Candidate empirical domains include climate mitigation, disaster preparedness, pandemic response, hunger hotspots, migration stress, and conflict spillovers. The goal is not to collapse them into one crude scalar. DAC should instead be treated as a family of measurable lags, conversions, and persistence relations that are comparable in structure rather than identical in raw form. Cross-domain comparability is therefore structured comparability across a DAC family.

A brief illustrative computation helps make the pathway concrete. In climate mitigation, one DAC relation can be defined as the lag and conversion strength between validated diagnostic escalation and fiscal actuation. A threshold event might be the publication of a recognized scientific warning or a nationally acknowledged carbon-budget overshoot. Translation can then be measured through the delay before the warning appears in binding policy language, the share of subsequent public expenditure that is explicitly aligned with the diagnostic, and the degree to which high-emissions subsidies or contradictory infrastructure investments remain in place. Deployment can be tracked through procurement, grid upgrades, permitting changes, or resilience investments, while persistence can be assessed by whether spending and implementation survive beyond the initial attention spike. The point is not that one ratio is sufficient, but that DAC can be computed as a structured family of warning-to-action relations rather than invoked as a metaphor.

A concrete worked example can be taken from pandemic response. On 30 January 2020, the World Health Organization declared the novel coronavirus outbreak a Public Health Emergency of International Concern, thereby providing a clear threshold event for institutional validation. Within a DAC framework, that declaration can be treated as t0 and compared with the subsequent timing of joint procurement, emergency financing, stockpiling authority, surge-capacity activation, and clinical-guidance dissemination across the European response architecture. The analytic question is not whether alarm existed, but how much of the validated signal survived translation into contracts, logistics, and durable coordination. Where procurement and deployment accelerated rapidly, DAC would register stronger coupling; where institutional acknowledgement outpaced supply, staffing, or distribution, DAC would remain weak despite high diagnostic visibility.

As an implementation pathway, EMO can be imagined as a layered sequence: interface registry; ingestion and harmonization; reciprocity and UIA engine; species-mind and planetary-health layer; DAC actuation layer; and governance interface. The actuation layer is where warning-to-policy lag, warning-to-budget lag, model-informed spending ratios, alert-to-deployment conversion, contradiction detection, and post-ignition persistence can be computed and compared. Even a modest v0.1 implementation would allow a first behavioral observatory rather than a purely diagnostic dashboard.

7. Implications for anticipatory governance and futures research

Reframing DAC as EMO's actuation layer gives the article a more explicit place within anticipatory governance. Futures research has long examined foresight, anticipation, visioning, and the uses of future-oriented knowledge. But the present challenge is no longer only whether future knowledge exists. It is whether future knowledge, once institutionally validated, can alter collective trajectories in time. DAC supplies a language for measuring that relation rather than presuming it. This placement also aligns the framework with established anticipatory-governance and futures scholarship, which has emphasized the difficult link between foresight, anticipatory systems, and present-day steering under uncertainty.

This also clarifies the paper's contribution to governance theory. The article is not simply proposing another grand metaphor of humanity. It is offering a way to study why sophisticated planetary diagnostics coexist with chronic under-response. In that sense, DAC is the missing response variable in a world that has rapidly expanded its sensing capacities. The more civilizational self-observation improves, the more the warning-to-action relation becomes the decisive empirical bottleneck.

For Futures specifically, the manuscript's relevance lies in joining three concerns often treated separately: the analysis of possible futures, the study of anticipatory systems, and the institutional problem of turning foresight into action. DAC makes that third concern measurable. EMO provides the infrastructure within which it can be observed. The combined architecture therefore speaks not only to philosophical questions about planetary cognition, but to practical questions about whether humanity can become behaviorally responsive at the scale of the dangers it now models.

Within foresight practice, this matters in at least three ways. First, EMO plus DAC allows scenarios to be compared not only by plausibility or desirability, but by likely actuation bottlenecks. A future may be analytically well described and still remain institutionally unreachable if warning-to-budget translation is chronically weak. Second, the framework provides a way to identify where anticipatory governance fails: not only in perception, but in the handoff from validated signal to organized delivery. Third, repeated measurement of DAC can help detect early behavioral mode transitions - for example, the shift from delayed coordination

to pulsed adaptation, or from selective stabilization to relapse - before those transitions become obvious in aggregate outcomes.

The practical value for futures research is therefore diagnostic as well as strategic. Foresight often succeeds in expanding the space of conceivable futures; DAC helps evaluate whether institutions possess the conversion pathways needed to move toward preferred ones. Used this way, the framework does not compete with scenario work, horizon scanning, or anticipatory governance. It adds a missing evaluative layer: whether the civilization's own future knowledge is behaviorally effective.

8. Limits, risks, and safeguards

This framework has limits. Coherence is not justice. Integration is not legitimacy. A system can become more coordinated without becoming more humane. EMO and DAC must therefore remain transparent, auditable, rights-preserving, and publicly contestable. Otherwise a diagnostic instrument becomes an instrument of domination.

The framework also does not eliminate contingency, leadership effects, culture, path dependence, or geopolitical asymmetry. It is not a deterministic machine for predicting the future. It is a structured way of narrowing expectation by identifying recurrent behavioral signatures of partial species integration under stress.

DAC in particular carries normative risks. A civilization may seek to strengthen actuation while weakening deliberation, dissent, or rights. Under those conditions, stronger coupling could become coercive rather than intelligent. This possibility must remain central to the theory. The goal is not maximal response compression at any cost, but higher-quality collective actuation under legitimate and contestable conditions.

A further limitation is comparative. DAC will almost certainly vary by regime type, state capacity, legitimacy structure, infrastructure quality, and historical memory. Identical diagnostic inputs should not be expected to yield identical actuation dynamics across liberal democracies, hybrid regimes, fragile states, or highly centralized systems. Cross-domain measurement will also face noise from uneven data quality, inconsistent temporal resolution, and proxy heterogeneity across sectors. DAC is therefore best treated as a patterned but context-sensitive family of relations.

9. Conclusion

The decisive question of this century may not be whether humanity can detect danger. It can. The deeper question is whether a species that has learned to sense at planetary scale can learn to act at planetary scale - and how strongly those two capacities are coupled.

The central move of this manuscript is to place diagnostic-to-actuation coupling inside the EMO architecture as its functional actuation layer. That placement strengthens the original argument. It clarifies novelty, reduces abstraction, and turns the warning-to-action gap into something more measurable than rhetorical frustration. EMO provides the observatory, self-model, and comparative ledger. DAC measures the conversion of that self-observation into organized response. Behavioral mode inference follows from their interaction.

The article's contribution is therefore methodological and anticipatory rather than metaphysical. It does not claim to prove the existence of a unitary species mind. It claims that humanity's conduct is becoming measurable enough to infer recurrent modes of response before full civilizational integration exists. The species becomes legible before it becomes integrated.

Put differently: the novelty of this framework is not species metaphor, but the measurable gap between diagnosis and actuation. The relevant threshold is the point at which that gap can be systematically observed, compared, and explained. EMO without DAC remains observational. DAC within EMO makes planetary self-observation behaviorally testable.

The next practical step is straightforward. Researchers can begin by operationalizing DAC in one or two high-salience domains, testing whether warning-to-action relations improve forecasting and identifying which sublayers - translation, deployment, persistence, or contradiction control - most strongly explain response failure. Institutions, in turn, can use EMO plus DAC not only to observe planetary stress, but to audit their own conversion capacity. In that narrower but more actionable sense, the framework offers a first pathway from planetary diagnostics to comparative response science.

Figure 1. Architectural logic of the framework

Planetary diagnostics and early warnings

↓

EMO observatory and shared self-model

↓

DAC actuation layer (validated warning -> translation -> budget -> deployment -> persistence)

↓

Behavioral mode inference -> Governance feedback and anticipatory adjustment

The figure is intentionally simple. Its purpose is not final formal closure, but to make the paper's central claim visually explicit: DAC is the actuation layer through which EMO's observatory architecture becomes behaviorally consequential.

Table 1. Core variables, operationalization, and candidate empirical proxies

Variable	Minimal operationalization	Candidate proxies
Organismality (O)	Degree of conflict suppression and cooperative continuity across scales under shared-risk conditions	Interstate conflict incidence; treaty compliance rates; arms-transfer volatility; cross-border emergency assistance; sanctions versus cooperation balance
Integrative processing (I)	Degree to which heterogeneous subsystems combine information into improved collective judgment	Cross-domain data fusion capacity; interagency model integration; multivariate synergy or O-information metrics; shared scientific-policy dashboards
Global workspace ignition (G)	Speed and scale with which signals become system-wide objects of attention	Synchronized spikes in news coverage, policy statements, web search behavior, institutional alerts, and scientific citations
Self-model fidelity (M)	Coupling between diagnostic representations and actual policy, budget,	Forecast-to-budget alignment; model-to-policy adoption rates; treaty-to-implementation lag; consistency between

	and delivery behavior	declared targets and infrastructure investment
Diagnostic-to-actuation coupling (D)	Strength with which validated warnings are translated into coherent, budgeted, and durable response	Warning-to-policy lag; warning-to-budget lag; alert-to-deployment conversion rates; continuity of implementation after attention spikes; model-informed spending ratios
Stress load (S)	Aggregate intensity of simultaneous ecological, geopolitical, metabolic, and informational pressures	Climate hazard frequency; displacement levels; hunger prevalence; energy and commodity shocks; conflict intensity; systemic volatility indices

Table 2. DAC within EMO: sublayers, measurement logic, and outputs

DAC sublayer	Question	Measurable relations	Illustrative outputs
Validation	Has a signal crossed the threshold from noise to recognized danger?	Model confidence; corroboration; institutional recognition; warning status	Validated risk event; recognized diagnostic; action-trigger eligibility
Translation	How much diagnostic content survives into policy and budget language?	Warning-to-policy lag; model-to-budget alignment; declared target/instrument consistency	Policy response score; budget translation ratio
Deployment	Does recognized danger become cross-scale operational delivery?	Alert-to-deployment conversion; logistics activation; cross-jurisdictional mobilization	Deployment index; implementation velocity
Persistence	Does response survive after attention spikes fade?	Continuity of spending; infrastructure follow-through; duration of coordination	Persistence coefficient; post-ignition retention
Contradiction and relapse	Do subsystems cancel one another out after apparent response?	Goal-budget mismatch; delivery reversal; contradictory incentives; decay after activation	Contradiction flag; relapse risk score

References

Author. (2025a). The human species as an emergent mind: From metaphor to measurement [Preprint]. Anonymized for review.

Author. (2025b). The Universal Interface Action: A cross-domain functional for curvature, coherence, and meaning [Preprint]. Anonymized for review.

Author. (2025c). The Emergent Mind Observatory: A UIA-based reciprocity ledger for species-level cognition and planetary health [Preprint]. Anonymized for review.

Clark, A., & Chalmers, D. (1998). The extended mind. Analysis, 58(1), 7-19. https://doi.org/10.1093/analys/58.1.7

European Commission. (2025). Destination Earth (DestinE): Digital model of the Earth. https://digital-strategy.ec.europa.eu/en/policies/destination-earth

Frank, A., Grinspoon, D., & Walker, S. I. (2022). Intelligence as a planetary-scale process. International Journal of Astrobiology, 21(2), 47-61. https://doi.org/10.1017/S147355042100029X

Mashour, G. A., Roelfsema, P., Changeux, J.-P., & Dehaene, S. (2020). Conscious processing and the global neuronal workspace hypothesis. Neuron, 105(5), 776-798. https://doi.org/10.1016/j.neuron.2020.01.026

OECD. (2019). Recommendation of the Council on Policy Coherence for Sustainable Development. OECD/LEGAL/0381. https://legalinstruments.oecd.org/en/instruments/oecd-legal-0381

OECD. (2024). Unleashing policy coherence to achieve the SDGs: An assessment of governance mechanisms. OECD Publishing. https://doi.org/10.1787/a1c8dbf8-en

Oh, A., Abazeed, A., & Chambers, D. A. (2021). Policy implementation science to advance population health: The potential for learning health policy systems. Frontiers in Public Health, 9, Article 681602. https://doi.org/10.3389/fpubh.2021.681602

Poli, R. (2010). An introduction to the ontology of anticipation. Futures, 42(7), 769-776. https://doi.org/10.1016/j.futures.2010.04.028

Priem, J., Piwowar, H., & Orr, R. (2022). OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2205.01833

Quay, R. (2010). Anticipatory governance: A tool for climate change adaptation. Journal of the American Planning Association, 76(4), 496-511. https://doi.org/10.1080/01944363.2010.508428

Queller, D. C., & Strassmann, J. E. (2009). Beyond society: The evolution of organismality. Philosophical Transactions of the Royal Society B: Biological Sciences, 364(1533), 3143-3155. https://doi.org/10.1098/rstb.2009.0095

Rockstrom, J., Gupta, J., Qin, D., et al. (2023). Safe and just Earth system boundaries. Nature, 619(7968), 102-111. https://doi.org/10.1038/s41586-023-06083-8

Siders, A. R. (2017). Adaptive governance for climate change: A systematic review. Wiley Interdisciplinary Reviews: Climate Change, 8(2), e463. https://doi.org/10.1002/wcc.463

United Nations Office for Disaster Risk Reduction. (2025). Early Warnings for All (EW4All). https://www.un.org/en/climatechange/early-warnings-for-all

United Nations Office for Disaster Risk Reduction. (2025). Global status of multi-hazard early warning systems 2025. UNDRR. https://www.undrr.org/reports/global-status-mhews-2025

Vervoort, J., & Gupta, A. (2018). Anticipating climate futures in a 1.5 °C era: The link between foresight and governance. Current Opinion in Environmental Sustainability, 31, 104-111. https://doi.org/10.1016/j.cosust.2018.01.004

World Health Organization. (2020). Statement on the second meeting of the International Health Regulations (2005) Emergency Committee regarding the outbreak of novel coronavirus (2019-nCoV), 30 January 2020. https://www.who.int/news/item/30-01-2020-statement-on-the-second-meeting-of-the-international-health-regulations-%282005%29-emergency-committee-regarding-the-outbreak-of-novel-coronavirus-%282019-ncov%29

European Union. (2020). Council Regulation (EU) 2020/521 of 14 April 2020 activating the emergency support under Regulation (EU) 2016/369, and amending its provisions taking into account the COVID-19 outbreak. EUR-Lex. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32020R0521

World Health Organization. (2025). Learnings from COVID-19 for future respiratory pathogen pandemic preparedness and response: A WHO technical package for implementation. https://iris.who.int/server/api/core/bitstreams/603334aa-1fa4-48fa-84cc-31a976efdfd3/content

World Health Organization. (2025). WHO Pandemic Agreement. https://www.who.int/health-topics/who-pandemic-agreement

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Research Data



Click here to download Research data https://github.com/avradri/emo-core-v1